Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Practice Worksheet

Chapter 1.3 (Storage & Compression) & Chapter 2.2, 2.3 (Transmission & Security)

Date: _____ **Student Name:** _____

Class: Secondary 2 (Sec 2) **Total Marks: 40 Marks | Obtained: _____**

SECTION 1: DATA STORAGE & FILE SIZE CALCULATIONS [10 MARKS]

1. Unit Conversions: (3 Marks)

Convert the following storage capacities into the requested units:

a) Convert 4 KiB into Bytes: _____

b) Convert 2.5 MiB into KiB: _____

c) Explain the difference between 1 Kilobyte (KB) and 1 Kibibyte (KiB):

_____

2. Image File Size Calculation: (3 Marks)

A bitmap image has a resolution of **1024 × 768 pixels** and a colour depth of **16 bits per pixel**.

Calculate the file size in **Kibibytes (KiB)**. Show your full working.

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3. Audio File Size Calculation: (4 Marks)

A studio records a **40-second stereo audio clip** with a sampling rate of **44,100 Hz** and a sample resolution of **16 bits**.

Calculate the size of the uncompressed audio file in **Bytes** and in **Mebibytes (MiB)**.

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